

Make My Figure ? Quick Start

From installation to an exported figure

MakeMyFigure version 1.0.0

Documentation generated from commit `ac49ff8`

Date 2026-09-03

Built from the current code and the running application. Screenshots and example figures use bundled synthetic data only.

Page references in the Contents count from the first body page.

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Make My Figure — Quick Start

MakeMyFigure version: 1.0.0

Documentation generated from commit: `ac49ff8`

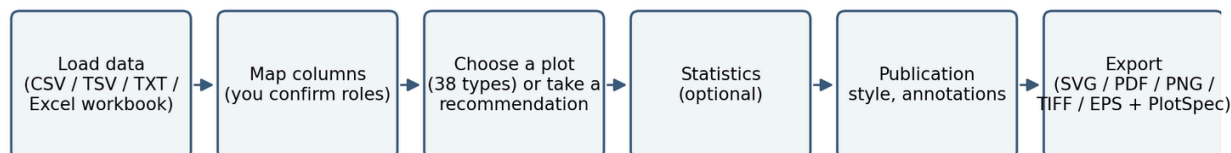
Date: 2026-09-03

This manual describes the application exactly as built at the commit above. Where the packaged installers of an earlier release differ, the text says so.

1. What Make My Figure does

Make My Figure turns a table you already have — a CSV, a TSV, or an Excel workbook — into a manuscript-style scientific figure, and records how the figure was made so it can be rebuilt later.

The path through the application is always the same:



Multi-panel assembly (Figure Builder) and Figure presets sit alongside every step.

Figure 1. The Make My Figure workflow: load, map columns, choose a plot (or take a recommendation), optional statistics, style and annotate, export with a PlotSpec record.

1. **Load** a table (or one worksheet of a workbook).
2. **Map columns** to roles — you confirm which column is x, y, group, p-value, time, event, and so on. The program never guesses biological meaning silently.
3. **Choose a plot type** from the 38 in the registry, or take one of the *advisory* recommendations.
4. **Optionally run statistics** — you choose the method; results are stored and every p-value drawn on the figure comes from a stored result.
5. **Style** the figure with the single **Publication** style (fonts, palette, layout, legend), add annotations.
6. **Assemble** several plots into a multi-panel figure if you need one.
7. **Export** SVG, PDF, PNG (and TIFF/EPS) together with a **PlotSpec** JSON that records the configuration.

Two interfaces share one engine: a **desktop app** (Qt) and a **browser app** (Streamlit). Both run entirely on your computer; nothing is uploaded anywhere.

Note: Make My Figure is a plotting and reporting tool. It computes the statistics you ask for and recommends plots from the *shape* of your table, but the choice of method, the interpretation, and journal compliance remain yours. Publication is a general manuscript style, not a journal template.

2. Installation

A. Packaged application (no Python needed)

Installers for **Windows** (MakeMyFigure-Setup.exe, or MakeMyFigure-windows.zip), **macOS** (MakeMyFigure.dmg) and **Linux** (MakeMyFigure.AppImage, or MakeMyFigure-linux.tar.gz) are attached to the GitHub release of version 1.0.0. They contain the desktop app and everything it needs.

Warning: at the time of writing the published installers were built from the v1.0.0 release (July 2026). They do **not** contain features added since — Figure presets, the histogram plot, the wide-form histogram picker, the collaborator fixes to survival curves and merged Excel headers. To use those today, install from source (B). A rebuilt installer will carry them once one is published.

- **Windows:** run the Setup .exe and follow the prompts, or unzip MakeMyFigure-windows.zip and run MakeMyFigure\MakeMyFigure.exe.
- **macOS:** open the .dmg and drag **Make My Figure** to Applications. The first launch may need right-click → **Open** because the bundle is not notarised.
- **Linux:** `chmod +x MakeMyFigure.AppImage && ./MakeMyFigure.AppImage`, or unpack the tarball and run `MakeMyFigure/MakeMyFigure`.

B. From source (desktop and browser apps)

Requirements: **Python 3.10–3.12** (the metadata allows 3.9, but Apple's bundled 3.9 failed pip's byte-compile step on a PySide6 file during testing — use a newer interpreter), git, and about 1 GB of disk for the Qt binding.

```
git clone https://github.com/surPoudel/make-my-figure.git
cd make-my-figure
python3 -m venv .venv
source .venv/bin/activate          # Windows PowerShell:
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install --no-compile -r requirements.txt
python -m pip install --no-compile -e ".[desktop]"      # adds PySide6 for the
desktop app
```

`--no-compile` skips a byte-compile step that fails on one PySide6 template file with some interpreters; Python compiles what it needs on first import.

Verify the build you are running:

```
python -c "from make_my_figure_core.version import build_banner;
print(build_banner())"
```

The line shows the version, the git commit, the platform and the Matplotlib backend. The same banner appears in the desktop status bar on launch and in the browser sidebar, so you can always confirm the manual matches the app.

Linux / WSL only: the Qt binding needs system libraries that minimal images lack:

```
sudo apt install -y libxkbcommon0 libxkbcommon-x11-0 libgl1 libegl1 libxcb-cursor0 \
  libxcb-xinerama0 libxcb-keysyms1 libxcb-randr0 libxcb-render-util0 libxcb-shape0 \
  libxcb-icccm4 libxcb-image0 libxcb-xfixes0 libdbus-1-3
```

If the desktop app cannot start it prints exactly this list. The browser app needs none of it.

3. Open the app

Desktop:

```
python -m apps.desktop_app.main
```

Browser:

```
streamlit run apps/streamlit_app/streamlit_app.py
```

and open the address Streamlit prints (normally `http://localhost:8501`).

The desktop app opens on the **Home / Upload** page. Drop a file onto it, choose **File** → **Open data file...**, or pick a bundled example from **File** → **Open example**.



Figure 2. Desktop app on launch: the Home / Upload page. Everything runs locally.

After data loads you are in the **plot workspace**: controls on the left (plot type, recommendations, column mapping, options, labels, Figure preset, Publication style, Statistics, Export), the data preview and messages at the top right, and the figure below them.

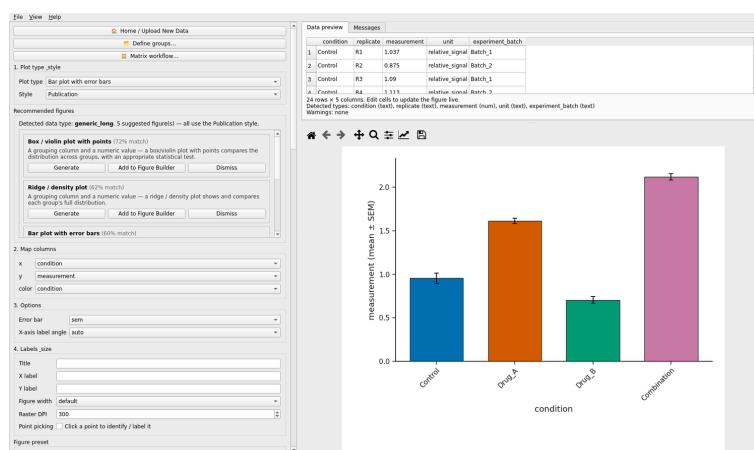


Figure 3. Desktop workspace with the bundled bar-plot example loaded: controls (left), data preview and messages (top)

right), live figure (bottom right).

The browser app has the same sections in its sidebar, numbered 1–6, with the table, the figure preview, Publication QC and Export in the main column.

Reset / Upload new data

Make My Figure v1.0.0 - ac49ff8
Linux - Agg

Diagnostics

1. Data

Data source

☒ Bundled sample

☐ Upload file

☐ Open PlotSpec

Example dataset (by plot type)

Bar plot with error bars

Synthetic example — Compare a mean readout across experimental conditions with replicate error bars.

Workflow

☒ Quick plot

☐ Matrix workflow (guided)

2. Plot type

Plot type

Bar plot with error bars

3. Style

Style

Publication

4. Column mapping

x

condition

y

measurement

color

condition

Error bar

sem

Figure preset

5. Publication style

Figure

Typography

Axes & labels

Legend

Colorbar

Not applicable to barplot with error bar: Style control 'marker_size' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar. Style control 'line_width_pt' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar. Style control 'legend_pt' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar. Style control 'legend_outside' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar.

6. Statistics

Statistical tests & annotations

Make My Figure

Manuscript-ready scientific figures in a single, polished Publication style. This is a general publication-ready visual style — not an official journal template, and not a guarantee of submission compliance.

Table — barplot_with_error_bar (example)

Edit cells directly to update the figure. Add/remove rows with the controls at the bottom of the table.

V_0 condition	V_0 replicate	V_0 measurement	V_0 unit	V_0 experiment_batch
Control	R1		1.037 relative_signal	Batch_1
Control	R2		0.875 relative_signal	Batch_1
Control	R3		1.09 relative_signal	Batch_1
Control	R4		1.113 relative_signal	Batch_2
Control	R5		0.766 relative_signal	Batch_1
Control	R6		0.844 relative_signal	Batch_2
Drug_A	R1		1.615 relative_signal	Batch_1
Drug_A	R2		1.562 relative_signal	Batch_1
Drug_A	R3		1.598 relative_signal	Batch_1
Drug_A	R4		1.498 relative_signal	Batch_2

Rows 24 Columns 5 Numeric columns 1

Define groups (no metadata file needed)

Recommended figures

Figure preview

Publication check: passed

Style control 'marker_size' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar.

Style control 'line_width_pt' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar.

Style control 'legend_pt' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar.

Style control 'legend_outside' bars have no markers or data lines; error-bar and edge widths come from the profile; it was ignored for this barplot_with_error_bar.

Publication QC — ready-to-export checks

Render metadata

Export

SVG PNG PDF PlotSpec JSON

Figure 4. Browser app with the same example: numbered sidebar sections, editable table, figure preview, export

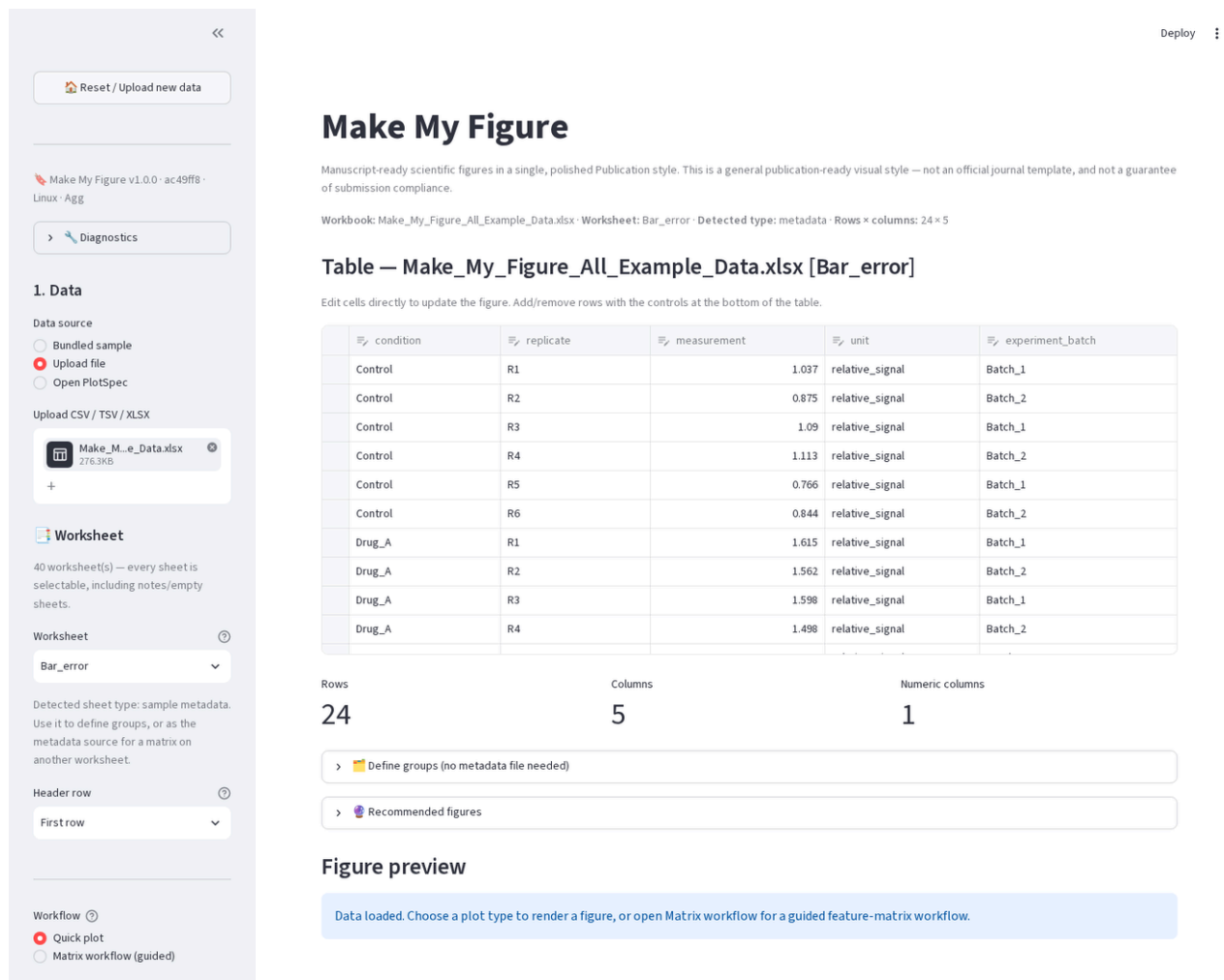
buttons.

4. Load data

Supported files: **CSV**, **TSV**, **TXT/TAB** (delimited text) and **Excel** **.xlsx**, **.xlsm**, **.xls**. Put column names in the first row and one observation per row. Numeric measurements should be numeric columns; identifiers and labels can be text.

- **Desktop**: drag a file onto the window, or **File** → **Open data file...**
- **Browser**: sidebar **1. Data** → **Upload file** → **Upload CSV / TSV / XLSX**

Excel workbooks with several sheets. Every worksheet is listed and selectable, including notes and empty sheets. Pick one in the  **Worksheet** box (desktop) or sidebar (browser). Each sheet is classified — *differential results*, *matrix*, *metadata*, *enrichment*, *documentation*, *generic*, *empty* — as a hint, not a decision; you can plot from any sheet.



Make My Figure

Manuscript-ready scientific figures in a single, polished Publication style. This is a general publication-ready visual style — not an official journal template, and not a guarantee of submission compliance.

Workbook: Make_My_Figure_All_Example_Data.xlsx · Worksheet: Bar_error · Detected type: metadata · Rows × columns: 24 × 5

Table — Make_My_Figure_All_Example_Data.xlsx [Bar_error]

Edit cells directly to update the figure. Add/remove rows with the controls at the bottom of the table.

condition	replicate	measurement	unit	experiment_batch
Control	R1	1.037	relative_signal	Batch_1
Control	R2	0.875	relative_signal	Batch_2
Control	R3	1.09	relative_signal	Batch_1
Control	R4	1.113	relative_signal	Batch_2
Control	R5	0.766	relative_signal	Batch_1
Control	R6	0.844	relative_signal	Batch_2
Drug_A	R1	1.615	relative_signal	Batch_1
Drug_A	R2	1.562	relative_signal	Batch_2
Drug_A	R3	1.598	relative_signal	Batch_1
Drug_A	R4	1.498	relative_signal	Batch_2

Rows: 24 Columns: 5 Numeric columns: 1

> Define groups (no metadata file needed)

> Recommended figures

Figure preview

Data loaded. Choose a plot type to render a figure, or open Matrix workflow for a guided feature-matrix workflow.

Figure 5. Browser app with the bundled example workbook: 40 worksheets listed, the selected sheet's detected type, and the header-row selector.

In the browser app you can also choose the **Header row** (first row, a chosen row, two stacked rows, or no header) and, for spreadsheets whose group label is written once over a block of replicate columns, tick **Carry a group label across blank cells**. Merged Excel cells are recovered automatically. The desktop app reads the first row as the header.

Matrix tables (features in rows, samples in columns, e.g. an expression or intensity matrix) work in the ordinary editor for heatmaps, clustering and PCA, and have their own guided **Matrix workflow**

(section 8).

5. Make a basic figure

Use the bundled example so the steps match the screenshots exactly: **File** → **Open example** → **Bar plot with error bars** (desktop) or **Bundled sample** → **Bar plot with error bars** (browser).

1. **Plot type** is already *Bar plot with error bars*; the **Style** is *Publication* (the only style).
2. **2. Map columns:** `x` → `condition`, `y` → `measurement`, `color` → `condition`. With your own data, set each role yourself — the defaults only prefill when a matching column exists.
3. **3. Options:** **Error bar** → `sem`, `sd`, `ci95` or `none`; **X-axis label angle**.
4. The figure re-renders as you change anything (the desktop debounces continuous controls so dragging stays responsive).
5. **4. Labels & size:** title, axis labels, **Figure width** preset (default / single / onehalf / double column) and **Raster DPI**.
6. **5. Publication style** — tick the box to enable it (desktop) or open the expanders (browser): typography, palette, tick angles, padding, legend location, colorbar, margins.

5. Publication style

② Typography

Palette (publication default)

Font (publication default)

Title pt 14

Axis label pt 12

Tick label pt 10

Annotation pt 10

Marker size 45

Line width 1.80

Axis/spine width 1.10

Grid ☐

③ Axes_labels

X tick angle auto

Y tick angle auto

X label padding 0.00

Y label padding 0.00

Title padding 0.00

④ Legend

Location auto

Legend pt 10

Legend outside ☐

⑤ Colorbar (heatmaps etc.)

Location default

Pad (0 = default) 0.00

Size 1.00

① Figure margins

Left (0 = auto) 0.00

Right (0 = auto) 0.00

Top (0 = auto) 0.00

Bottom (0 = auto) 0.00

Auto-fix layout ☐

Reset to publication defaults

Figure 6. The Publication style group: ② Typography, ③ Axes and labels, ④ Legend, ⑤ Colorbar, ① Figure margins,

and Reset to publication defaults.

1. **Export:** desktop **5. Export** → **Export SVG / PNG / PDF / PlotSpec JSON** or **Export all as ZIP**; browser **Export** → **SVG / PNG / PDF / PlotSpec JSON** download buttons. A `.plot_spec.json` sidecar is written with every export so the figure can be reproduced.

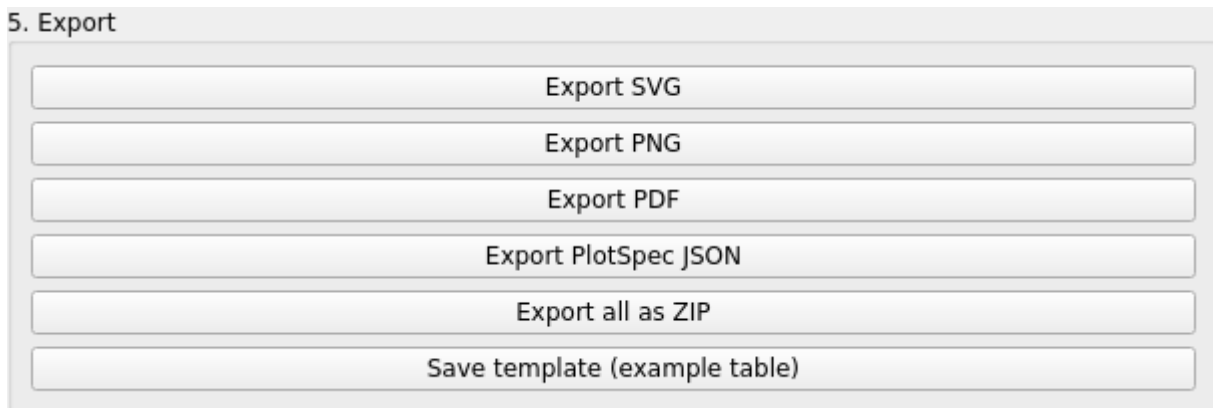


Figure 7. Desktop Export group: SVG, PNG, PDF, PlotSpec JSON, ZIP bundle, template table, and the multi-panel buttons.

Controls that do not apply to the current plot are reported rather than silently ignored — for a bar plot, for example, the marker-size and line-width controls do nothing and the app says so.

6. Use recommendations

When a table loads, Make My Figure profiles it (numeric vs. categorical columns, identifiers, p-value-like columns, matrix shape, edge lists, survival columns...) and suggests plots with a match score and the reason.

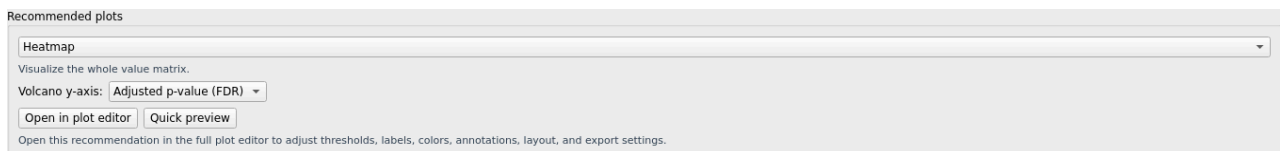


Figure 8. Recommended plots for a grouped-observation table: each card explains the match and offers Generate, Add to Figure Builder, or Dismiss.

- **Desktop:** the **Recommended figures** group under the plot type; **Generate** switches to that plot with the suggested mapping; **Add to Figure Builder** saves it as a panel.
- **Browser:** the  **Recommended figures** expander above the figure.

Recommendations are **advisory**. The score is a rule-based heuristic on the table's shape, not a measure of scientific appropriateness. You remain responsible for choosing what the figure should show.

7. Run statistics

Example: **Box / violin plot with points** (bundled). In **6. Statistics** (desktop: tick the group box, then **Enable statistics**; browser: **Statistical tests & annotations** expander):

1. **Test** — e.g. *Welch's t-test*. Eighteen methods are available (t-tests, Mann-Whitney, Wilcoxon, ANOVA variants, Kruskal-Wallis, Dunn, chi-square, Fisher, log-rank, Cox, Pearson, Spearman, linear regression, GLM). *Auto-suggest* proposes one from the mapping; it is a suggestion, not a decision.
2. **Comparison** — all groups pairwise, compare to a control, within each x category, selected

pairs, or omnibus only.

3. **Correction** — Benjamini–Hochberg FDR, Bonferroni, Holm, or none.
4. **Annotation shows / placement** — stars, exact p, adjusted p, statistic, effect size...; brackets between compared groups, or one label above each bar when comparing to a control.
5. **Run statistics.** The results table lists each comparison with p, adjusted p, effect size and n, and a **method sentence** is written for your Methods section. Export the table (CSV) or the method report (Markdown).

☒ 6. Statistics

☒ Enable statistics

Test

Welch's t-test

Comparison

Compare all groups (pairwise)

Group column

(none)

Subgroup column

(none)

Subject/pair ID

(none)

Control group

(none)

Correction

Benjamini-Hochberg FDR

Annotation shows

Stars only

Annotation placement

Bracket between the compared bars

Custom template

☐ Show effect size on figure

☐ Hide non-significant annotations

☐ Post-hoc pairwise after omnibus

P-value decimals

3

Annotation font pt

9.50

Run statistics

Clear

Reset

Suggested: One-way ANOVA. Three or more groups: one-way ANOVA or Kruskal-Wallis, optionally with post-hoc pairwise tests and correction.

	Comparison	Test	p	adj p	effect	n
1	Healthy vs Disease	Welch's t-test	5.971e-12	1.791e-11	Hedges' g=-2.67	48
2	Healthy vs Treated	Welch's t-test	3.051e-05	4.577e-05	Hedges' g=-1.33	48
3	Disease vs Treated	Welch's t-test	0.0006929	0.0006929	Hedges' g=1.03	48

Statistical analyses were performed with Python (scipy 1.17.1, statsmodels 0.14.6, numpy 2.1.2). Tests used: Welch's t-test. P-values were adjusted across 3 comparisons using Benjamini-Hochberg FDR correction. Exact p-values are reported unless otherwise noted. Non-finite values were removed listwise before testing. Users are responsible for confirming that each test is appropriate for their experimental design.

Export stats table

Export method report

Copy method sentence

Figure 9. The Statistics panel after Run statistics on the box/violin example: Welch's t-test, all pairs, BH correction,

Warning: the program performs the calculation you request. It does not verify that a test suits your design. Check assumptions (independence, pairing, distribution) before you rely on a result.

8. Matrix workflow

For a feature-by-sample matrix, open **Matrix workflow...** (desktop) or choose **Workflow → Matrix workflow (guided)** (browser).

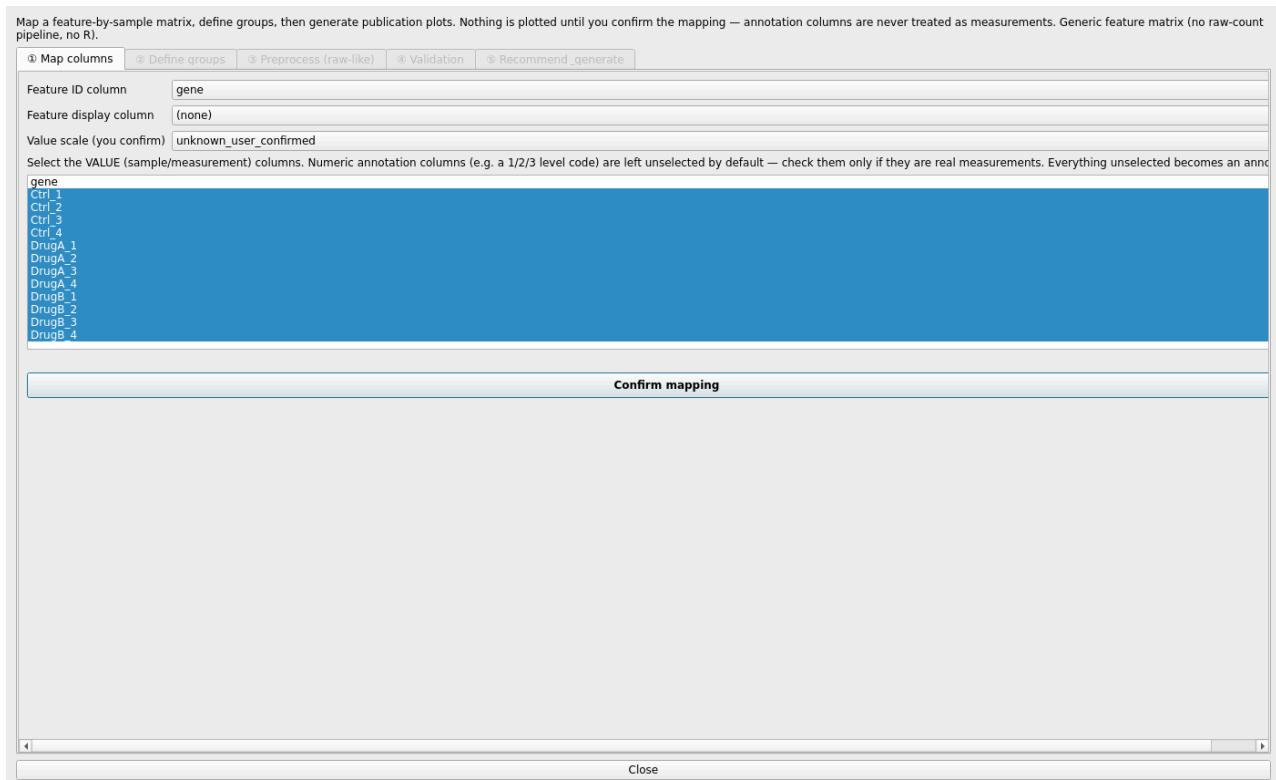


Figure 10. Matrix workflow, step ① Map columns: feature ID column, value columns, and the value scale you confirm before anything is plotted.

1. ① **Map columns** — the feature ID column, optional display column, the **value (sample) columns**, and the **value scale** (normalized, log_normalized, raw_numeric, or unknown_user_confirmed). Numeric annotation columns are left unselected until you tick them. Click **Confirm mapping**.
2. ② **Define groups** — upload a metadata file, take one from another worksheet, or type a group per sample column; then **Confirm groups**.
3. ③ **Preprocess (raw-like)** — optional, for raw-like data: **Run diagnostics** (suspected data type, skew, zeros, negatives), preview a **QC plot**, pick one of the **recommended preprocessing** recipes (e.g. *Total-sum normalize → log2 → z-score*; nothing is applied automatically), then **Apply preprocessing → use processed matrix**. Each step is recorded; the original matrix is never modified; **Save before/after QC report...** writes the report.
4. ④ **Validation** — a report of the confirmed mapping and groups (features, samples, groups, missing values, duplicate ids) with warnings or ✓ *Validation passed*.
5. ⑤ **Recommend & generate** — an optional two-group **differential summary** (test, correction, groups A/B), then recommended plots (heatmap, PCA, box plots, volcano/MA once a summary exists) with **Quick preview** and **Open in plot editor**, which hands the derived table to the full editor with its provenance.

9. Save a Figure preset

A **Figure preset** is your configuration without your data. Configure a figure once, save the preset, and apply it to a new dataset of the same plot type later.

- **Figure style only** — fonts, palette, line/marker sizes, tick angles, margins, legend and colorbar placement, export size/DPI, and visual plot options (violin vs box, colormap, node colours...). Portable: this is the lab's *Lab_Default_Heatmap.mmfpreset.json*.
- **Full figure configuration** — everything above plus column roles, thresholds, axis labels and the statistics test. On new data it applies what fits and lists the columns it could not find, for you to map — it never guesses.

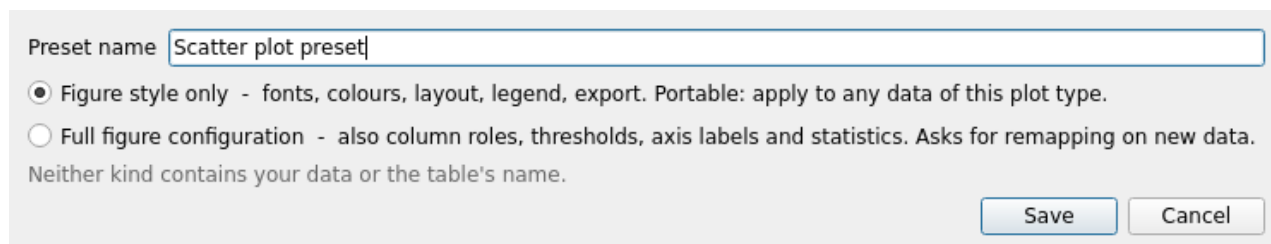


Figure 11. Save Figure Preset: name it and choose style-only or full configuration. Neither kind contains your data.

Desktop: the **Figure preset** group (above Publication style) — choose a preset, **Apply**, **Save preset...**, **Delete**, **Import...**, **Export...**, **Reset to Publication defaults**; the same actions are in **File** → **Figure preset**. **Browser:** the **Figure preset** expander in the sidebar.

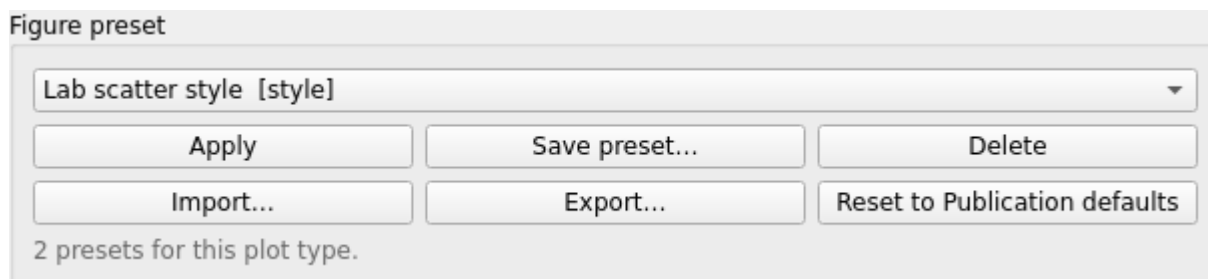


Figure 12. The Figure preset group with saved presets listed for the current plot type.

A **PlotSpec** is different: it is the exact record of *one* figure, including the table name and column names. Every export writes one. An existing PlotSpec can be loaded as a full preset.

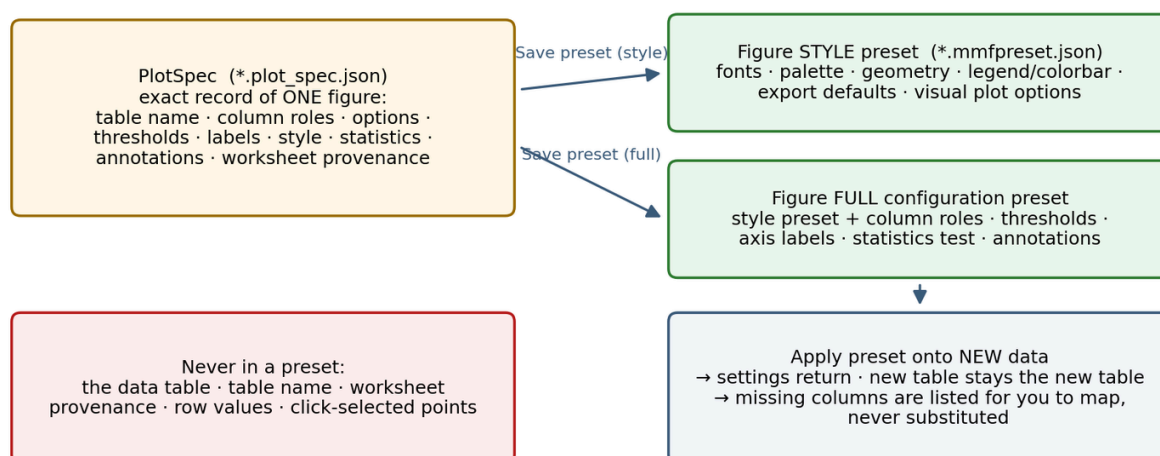


Figure 13. PlotSpec versus Figure preset: what each contains, and what never enters a preset.

10. Figure Builder (multi-panel figures)

In the desktop app, click **Save current plot as panel** for each plot you want, then **Open Figure Builder....**

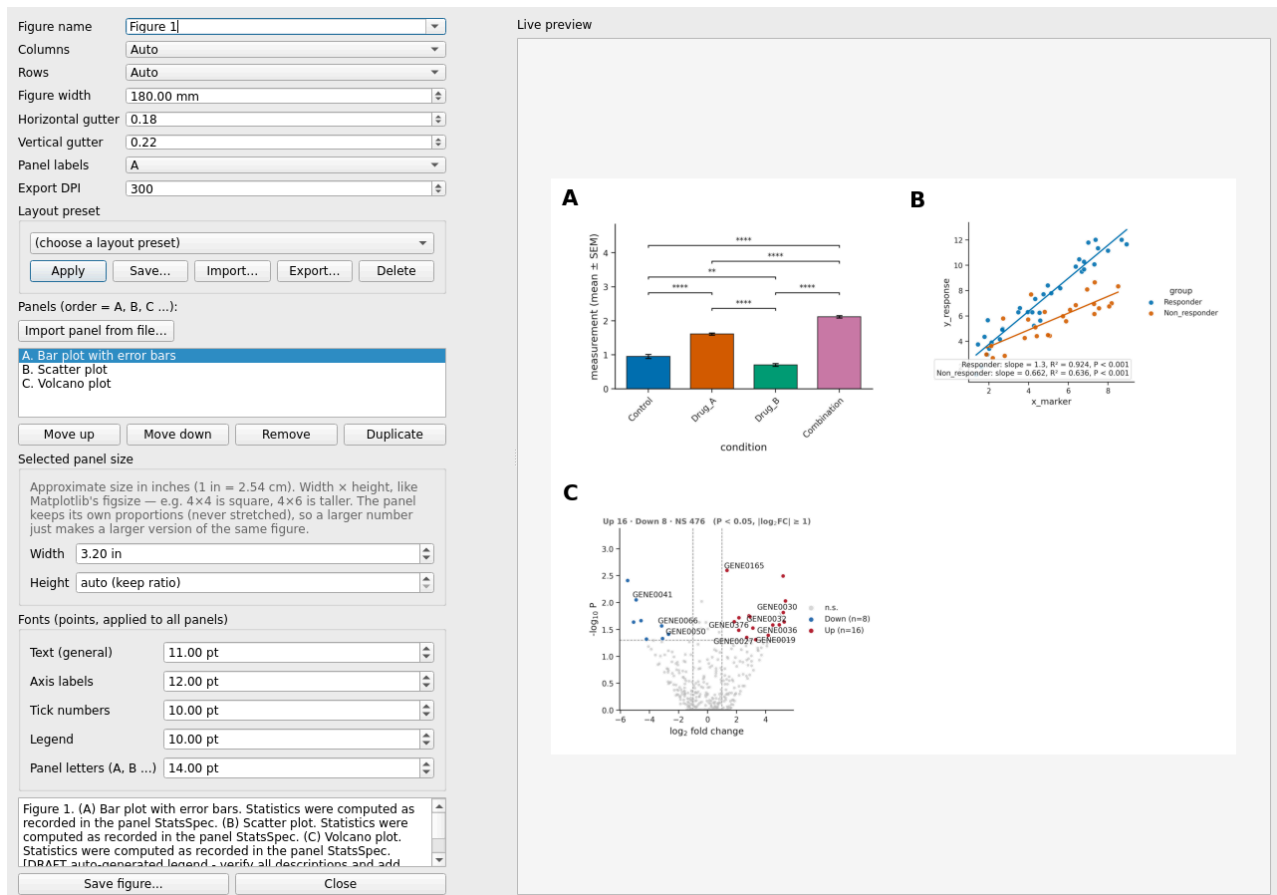


Figure 14. Figure Builder with three saved panels: grid, gutters, panel-letter style, per-panel size in inches, global fonts, layout presets, and a live preview.

- Add generated plots (they re-render from their PlotSpecs) or **Import panel from file...** (PNG, JPG, TIFF, WEBP, BMP; PDF/SVG/EPS with the optional `import-panels` extra).
- Set **Columns**, **Rows**, **Figure width** (mm), the two **gutters**, **Panel labels** (A / a / 1) and **Export DPI**.
- Select a panel to set its **Width** and **Height** in inches (proportions are preserved, never stretched), reorder, duplicate or remove it.
- **Fonts** apply to every panel. **Layout preset** saves the grid/sizes/fonts for reuse without any panel content.
- **Save figure...** writes PNG/SVG/PDF plus a **FigureSpec** (`.figure_spec.json`).

Note: the composite embeds each panel's content as a raster image at the export DPI; panel letters and titles stay vector text. For a fully vector single panel, export that plot directly as SVG or PDF. The browser app has a simpler Figure Builder (columns only) inside the guided Matrix workflow.

11. Export

Format	Type	DPI	Text	Typical use
--------	------	-----	------	-------------

SVG	vector	—	editable	edit further in Illustrator/Inkscape; journals accepting SVG
PDF	vector	—	editable (embedded Type 42 fonts)	manuscript submission, print
EPS	vector	—	editable	legacy journal systems (core/ZIP export)
PNG	raster	150–600 (set in Raster DPI)	—	slides, previews, journals requiring raster
TIFF	raster (LZW)	as PNG	—	journals requiring TIFF (core/ZIP export)

Every export also writes `name.plot_spec.json`; when statistics ran, `name.stats_spec.json` as well. The desktop **Export all as ZIP** bundles formats and sidecars.

12. Where to get help

- **Desktop Help menu: Help...** (plot types, how to format your data, privacy, the Publication-style disclaimer), **About** (version), **Copy debug info**.
- **Browser sidebar:** the version banner and 🐞 **Diagnostics**.
- The full **User Manual** (this folder) documents every control, plot, statistic and file.
- **Repository:** <https://github.com/surPoudel/make-my-figure> (issues, if the repository is public for you).

Always quote the banner line (version · commit · platform · backend) when reporting a problem.